PMA^2-Net: Progressive Fusion of Multiscale Axial Attention Network for Hyperspectral and Multispectral Images

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***Abstract*— Integrating low-resolution hyperspectral images (LR-HSIs) with corresponding high-resolution multispectral images (HR-MSIs) for the reconstruction of HR-HSI using deep learning techniques represents a critical area of research. Although convolutional neural networks (CNNs) are widely used for HR-HSI reconstruction, their small receptive fields hinder effective global feature extraction, which limits their potential. Fusion methods that rely on traditional attention mechanisms also lack feature interaction, failing to integrate and harmonize feature information from hyperspectral image (HSI) and MSI effectively. To address these issues, this article develops a novel progressive fusion of multiscale axial attention network (PMA^2-Net), which combines multiscale convolutions with axial attention (AA) and employs a progressive interaction approach to reconstruct high-resolution images. Specifically, PMA^2-Net extracts the spatial and spectral information from HSI through spatial feature extraction (Spatial-FE) and spectral feature extraction (Spectral-FE). Concurrently, a feature injection module (FIM) is introduced, employing multiscale convolution to capture local features and integrating AA to enhance global feature association. Moreover, a progressive fusion module (PFM) is employed to enhance multidimensional feature collaboration and hierarchical integration. Extensive studies conducted using five significant HSI datasets verify the effectiveness of PMA^2-Net, demonstrating its superior performance compared to current state-of-the-art (SOTA) fusion techniques.**

***Index Terms*— Axial attention (AA), hyperspectral image (HSI), image fusion, multispectral image (MSI).**

I. Introduction

HYPERSPECTRAL images (HSIs) can accurately reflect differences in ground objects through hundreds of narrow, consecutive spectral bands. They provide significant advantages in tasks such as image classification [1], [2], object detection [3], [4], [5], and change detection [6], [7], [8]. However, the spatial resolution of HSIs is relatively low to maintain an acceptable signal-to-noise ratio. Conversely, multispectral images provide high spatial resolution but insufficient spectral resolution. To address this contradiction, image fusion techniques have been developed to combine low-resolution HSI (LR-HSI) with high-resolution multispectral image (HR-MSI) to reconstruct HR-HSI, offering accurate, comprehensive information for various tasks. These fused images are valuable for applications such as fine crop classification [9], mineral exploration [10], and land cover classification [11], [12].

Image fusion technology has been developed over a long period. Pansharpening wavelet techniques inspired early methods for HSI and MSI image fusion [13]. In applying these MSI and HSI fusion methods, specific MSI bands are selected to enhance corresponding HSI bands. HSI and MSI can also be represented as a 3-D tensor. Based on this, numerous approaches have been proposed, including matrix decomposition-based methods [14], subspace representation [15], and tensor factorization [16], [17]. Most of these methods rely on manually designed prior knowledge and fusion rules, making them time-consuming, limited in expressive power, and unable to adapt to complex fusion scenarios.

Given the swift advancements in deep learning technology, convolutional neural networks (CNNs) have proven to be exceptionally effective in the realm of computer vision. CNN-based methods demonstrate remarkable expressive power and effectiveness when combining LR-HSI and HR-MSI. SSR-NET [18] effectively fuses HSI and MSI modalities using spatial–spectral edge loss, significantly enhancing the quality of fusion images. Addressing the challenge of matching the resolution of HSI and MSI, MCT-NET [19] integrates the spatial–spectral characteristics of HSI and MSI using a multilevel cross-transformer (MCT) and stripe convolution structure. To improve the fidelity of spectral and spatial features in HSI fusion, HSR-Net [20] integrates channel attention (CA) and spatial attention (SA) modules. Additionally, it ensures the preservation of fine structures during the upsampling process through the PixelShuffle module. SS-INR [21] is pioneering in its application of implicit neural representations (INRs) technology to HSI hyperspectral resolution reconstruction, compared to traditional upsampling methods that often inadequately recover continuous

Received 21 July 2024; revised 29 October 2024; accepted 7 December 2024. Date of publication 11 December 2024; date of current version 19 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42171388 and Grant 41925006. *(Corresponding authors: Yuebin Wang; Danfeng Hong.)*

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Digital Object Identifier 10.1109/TGRS.2024.3514893

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spatial–spectral information. This method constructs a fusion framework based on a deep backprojection network, using the degradation process to guide network optimization, leveraging inherent correlations between high- and low-resolution images, and significantly improving fusion image quality.

Due to the limited receptive field of convolutional kernels, CNN-based fusion methods struggle to capture the full range of global relationships within feature maps [22]. Local features lack the capacity to represent global context, multiscale dependencies, and complex structures, making it challenging to meet the demands of high-resolution HSI reconstruction, which requires both global information and high-frequency details.

Transformer-based methods [19], [22], [23], [24], which capitalize on self-attention's strong ability to capture long-range dependencies [25], have been widely applied in MSI and HSI fusion. However, the quadratic computational complexity of self-attention results in considerable computational and memory overhead when handling high-resolution images or high-dimensional data, thereby reducing feature extraction efficiency, particularly in multimodal information fusion tasks. In contrast, axial attention (AA) reduces complexity by applying attention mechanisms separately along rows and columns [26], enhancing computational efficiency and memory utilization while effectively capturing global features. However, it falls short in multiscale feature extraction, as it struggles to integrate detailed information from different scales simultaneously. This limitation results in incomplete representation of high-resolution details, ultimately affecting the effectiveness of feature extraction and fusion. To address these issues, we propose a novel progressive fusion of multiscale axial attention network (PMA^2-Net) for fusing LR-HSI and HR-MSI.

The network adopts a two-branch architecture that separately extracts HSI spatial and spectral features, referred to as spatial feature extraction (Spatial-FE) and spectral feature extraction (Spectral-FE), respectively. Subsequently, we design a feature injection module (FIM) to efficiently extract multiscale local and global features from MSI. Specifically, the multiscale AA (MAA) Module enables each pixel to capture dependencies across the entire image. The extracted features are hierarchically injected into the feature fusion process, enabling different data sources to share contextual information. Additionally, we design a progressive fusion module (PFM). The PFM gradually maps low-resolution features to high-resolution space, continually refining and optimizing the reconstruction results with injected spatial information from each layer. This layer-by-layer fusion process ensures that the model not only restores high-frequency details of HSI but also preserves and integrates multiscale features, effectively enhancing the spatial quality and integrity of the reconstructed images.

The following is a summary of this article's primary contributions.

1) In this study, we innovatively design PMA^2-Net, a deep learning network dedicated to the fusion of HSI and MSI. PMA^2-Net combines a dual-branch feature extraction module with a FIM alongside a PFM. The efficacy and superiority of the suggested PMA^2-Net in fusing LR-HSI and HR-MSI are consistently confirmed by experimental data.
2) We design a FIM, which innovatively harnesses MAA to extract multiscale local–global features from MSI, incorporating hierarchical downsampling to enhance detail injection in HR-HSI reconstruction, preserving object accuracy and complexity in high-resolution scenarios.
3) We propose the PFM, which innovatively maps low-resolution features progressively to high-resolution spaces and integrates multilayer spatial information to optimize reconstruction, enabling both restoration of high-frequency details and integration of multiscale features.

The remainder of this article is organized as follows. Section II reviews related work with a particular focus on existing fusion methods. Section III comprehensively describes the proposed PMA^2-Net model. Section IV discusses the experimental results in detail, comparing the performance of the PMA^2-Net model against 11 state-of-the-art (SOTA) algorithms across five hyperspectral datasets. Additionally, it includes results from an ablation study. Finally, Section V provides the concluding remarks.

II. Related Work

This section introduces some current traditional techniques for fusing MSI and HSI, such as Pansharpening, Matrix Factorization, and Tensor Factorization. Next, we give a summary of deep learning-based fusion techniques.

A. Classical Fusion Methods

Pansharpening was initially employed to enhance the spatial resolution of MSI, achieving or surpassing the clarity of panchromatic (Pan) images. Most early HSI and MSI fusion methods, such as component substitution (CS) and multiresolution analysis (MRA), originate from pansharpening [27]. Expanding on established pansharpening techniques, Chen et al. [28] presented a novel framework for combining HSI and MSI. Picone et al. [29] introduced a band assignment method based on the spectral angle mapper (SAM) criterion to minimize spectral distortion. Departing from conventional band allocation methods, Selva et al. [30] synthesized high-resolution counterparts for each hyperspectral band using a regression framework with MSI.

Methods based on matrix factorization transform LR-HSI and HR-MSI into 2-D representations, subsequently decomposing them into spectral bases and a matrix of coefficients [31]. Using the sparse matrix factorization (SMF) approach, Kawakami et al. [32] estimated the basis matrix with a decomposition algorithm that leverages the spatial sparsity of HSIs. To update the bases and coefficients alternately, Yokoya et al. [33] proposed using coupled nonnegative matrix factorization (CNMF). Simoes et al. [15] employed vertex component analysis (VCA) to learn the bases within a low-rank subspace.

In tensor factorization-based methods, two commonly employed approaches are Tucker Factorization and canonical

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Polyadic (CP) decomposition [22]. Li et al. [16] proposed a coupled sparse tensor factorization (CSTF) scheme, redefining the fusion problem as an estimation of the core tensor and three-mode dictionaries with a regularization term to promote sparse core tensors. Using a CP decomposition approach, Kanatsoulis et al. [34] proposed a coupled tensor decomposition framework that leverages the multidimensional structure of HSI and MSI to factorize HR-HSI.

B. Deep Learning Fusion Methods

Recently, image fusion approaches have advanced significantly, driven by the rapid expansion of deep learning technologies. The efficient feature extraction and precise reconstruction capabilities of neural networks have driven remarkable progress in image fusion, with an increasing number of deep learning methods employed. Palsson et al. [35] were the first to propose a 3-D CNN for fusing MSI and HSI to generate HR-HSI. Since most current HSI enhancement techniques rely on image prior modeling, often sensitive to parameter choices and time-intensive, Dian et al. [36] proposed a deep hyperspectral enhancement method (DHSIS). By utilizing deep residual learning priors, they effectively regularized the image fusion problem. Han et al. [37] introduced two innovative CNN architectures, namely SSFCNN and ConSSFCNN, effectively leveraging complementary information from dual-source images to achieve super-resolution reconstruction of HR-HSI.

The fusion process of LR-HSI and HR-MSI involves constructing a complex mapping model for transforming spatial resolution from low to high. Inspired by the strong representational capabilities of deep neural networks in capturing nonlinear relationships, Yuan et al. [38] proposed a remote sensing image fusion CNN named MSDCNN. Liu et al. [39] proposed a two-stream fusion network (TFNet) for pan-sharpening, which fuses Pan and MSI in the feature domain using separate CNNs for spatial and Spectral-FE. Sharp edges and low-level structural information are frequently lost or diminished as neural networks deepen. To address this, Zheng et al. [40] presented the edge-conditioned feature transformation network (EC-FTN), intended to maintain low-level structural information, like sharp edges. However, more than the traditional fusion of HSI and MSI alone, it is necessary to enhance spatial details fully. Therefore, Li et al. [41] introduced HyperNet. This unified fusion framework simultaneously processes three image sources (HSI, MSI, and PAN), effectively addressing the information loss issues associated with previous staged fusion approaches.

Despite the success of CNNs in HSI super-resolution reconstruction, they are limited by local receptive fields and overlook global correlations. Fusformer [42] innovatively combines the Transformer architecture to overcome CNN limitations and achieves global feature correlation modeling. It demonstrates remarkable spectral image recovery results with a smaller parameter count and exhibits strong generalization ability. Chen et al. [43] proposed the spectral–spatial transformer (SST) framework, which has a spectral–spatial dual-branch structure specifically designed for HS and MS images. The Transformer module within this framework effectively captures long-range dependencies between HS and MS images, addressing the low spatial resolution issue due to sensor constraints in HSI and confirming the significant potential of Transformers in enhancing HSI quality. Wang et al. [23] introduced the convolution and contextual transformer (CCoT) method, which combines the inductive bias of CNNs and the powerful global modeling of Transformers to acquire spectral features, thereby improving reconstruction quality more effectively. MSST-Net [22] incorporates the self-attention mechanism and employs a self-supervised pretraining strategy to boost performance. Fang et al. [44] proposed a multi-input multi-output SST (MIMO-SST) for HSI and MSI fusion. MIMO-SST integrates spatial–spectral information with multihead attention and a multiscale CNN, while a wavelet-based loss enhances edge detail recovery.

III. Proposed Method

This section provides a detailed introduction to the proposed PMA^2-Net. Fig. 1 illustrates the overall architecture of PMA^2-Net. The key components of the PMA^2-Net architecture include four modules: 1) Spatial-FE; 2) Spectral-FE; 3) FIM; and 4) PFM.

A. Spatial-FE-HSI

HSIs integrate rich spectral information, complex spatial textures, and morphological structures, often making it difficult to separate these intertwined features directly. However, traditional image processing techniques frequently fail to accurately capture these crucial spatial features. To address this challenge, we design a Spatial-FE module, with the process illustrated in Fig. 1(a).

1) Axial Attention: AA effectively integrates attention mechanisms into image processing by performing self-attention operations along the rows and columns of the input feature maps, enabling global information modeling while significantly reducing computational complexity. As depicted in Fig. 2, the specific formula can be represented as

$$\tilde{X}_{\text{out}} = \text{ColAtten}(F_{\text{Colde}}(\text{RowAtten}(F_{\text{Rowde}}(X_{\text{in}})))) \quad (1)$$

where $X_{\text{in}} \in \mathbb{R}^{m\times n\times 1}$ is first decomposed along the rows using $F_{\text{Rowde}}(\cdot)$, resulting in $X_{\text{row}} = \left[r_1\ r_2\ \cdots\ r_m\right]^{\text{T}}$, where each $r_i \in \mathbb{R}^{n\times 1}$. Then, X_{row} undergoes row attention RowAtten$(\cdot)$, yielding $\tilde{X}_{\text{row}} \in \mathbb{R}^{m\times n\times 1}$. The specific formula is as follows:

$$(Q_{\text{row}}, K_{\text{row}}, V_{\text{row}}) = X_{\text{row}} \times \left(W_{Q_{\text{row}}}, W_{k_{\text{row}}}, W_{V_{\text{row}}}\right) \quad (2)$$

$$\tilde{X}_{\text{row}} = \text{softmax}\left(\frac{Q_{\text{row}}K_{\text{row}}^{\text{T}}}{\sqrt{n}}\right)V_{\text{row}} \quad (3)$$

where $W_{Q_{\text{row}}}$, $W_{K_{\text{row}}}$, and $W_{V_{\text{row}}}$ are the projection matrices for the query, key, and value, respectively, each with a shape of $n \times n$. The corresponding outputs Q_{row}, K_{row}, and V_{row} all have the shape $m \times n$. After row attention, $\tilde{X}_{\text{row}}$ undergoes column decomposition using $F_{\text{Colde}}(\cdot)$, to yield $X_{\text{col}} = \left[c_1\ c_2\ \cdots\ c_n\right]$, where each $c_i \in \mathbb{R}^{m\times 1}$. Similar to row attention,

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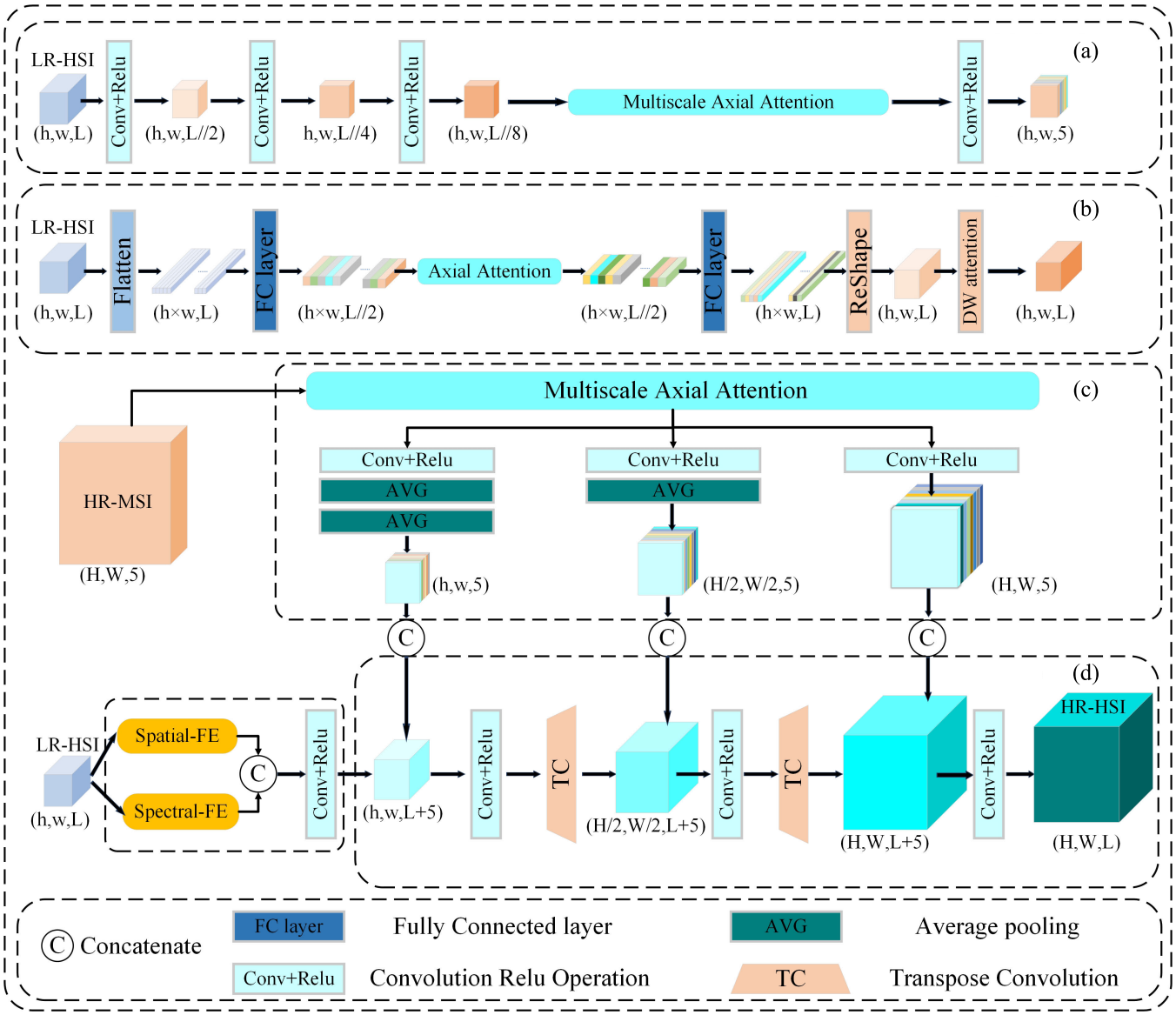


Fig. 1. Overall structure diagram of the proposed PMA^2-Net. (a) Spatial-FE. (b) Spectral-FE. (c) FIM. (d) PFM. HSI is processed through two independent modules, Spatial-FE and Spectral-FE, which refine spatial and spectral features separately. MSI utilizes the FIM to capture local–global features at different scales, integrating them into the reconstruction process through downsampling. The PFM progressively maps low-resolution features to high-resolution spaces, achieving high-quality reconstruction of HR-HSI.

column attention ColAtten(·), is applied, with the specific formula as follows:

$$(Q_{\text{col}}, K_{\text{col}}, V_{\text{col}}) = X_{\text{col}} \times \left(W_{Q_{\text{col}}}, W_{k_{\text{col}}}, W_{V_{\text{col}}}\right) \tag{4}$$

$$\tilde{X}_{\text{col}} = \text{softmax}\left(\frac{Q_{\text{col}} K_{\text{col}}^{\text{T}}}{\sqrt{m}}\right) V_{\text{col}} \tag{5}$$

where $W_{Q_{\text{col}}}$, $W_{K_{\text{col}}}$, and $W_{V_{\text{col}}}$ are the projection matrices corresponding to the column-wise Query, Key, and Value, each with a shape of $m \times m$. The resulting outputs Q_{col}, K_{col}, and V_{col} all have shapes of $n \times m$. The aforementioned computational process can be succinctly represented as

$$\tilde{X}_{\text{out}} = \text{AA}(X_{\text{in}}) \tag{6}$$

where AA efficiently extracts local–global features $\tilde{X}_{\text{out}} \in \mathbb{R}^{m\times n\times 1}$ from the input feature map $X_{\text{in}} \in \mathbb{R}^{m\times n\times 1}$.

2) MAA Module: The MAA Module effectively extracts local–global features through the combination of multiscale convolution and AA. Fig. 3 illustrates the particular procedure. The formula in question can be expressed as follows:

$$\begin{aligned} \tilde{X}_i &= (\text{AA}(\text{Conv}_i(X_{\text{IN}})) + \text{Conv}_i(X_{\text{IN}}))_{i\in 1,3,5,7,9} \\ \tilde{X} &= \text{Concat}\left(\{\tilde{X}_i\}_{i\in\{1,3,5,7,9\}}\right) \end{aligned} \tag{7}$$

where $\text{Conv}_i(\cdot)$, represents a convolutional layer with a kernel size of i and Concat(·), denotes concatenation. The input feature map $X_{\text{IN}} \in \mathbb{R}^{m\times n\times c}$ acquires multiscale local and global feature information $\tilde{X} \in \mathbb{R}^{m\times n\times 5}$ after passing through the MAA. This computational process can be succinctly represented as

$$\tilde{X} = F_{\text{MAA}}(X_{\text{IN}}). \tag{8}$$

3) Spatial Feature Extraction: As shown in Fig. 1(a), LR-HSI (lr $\in \mathbb{R}^{h\times w\times L}$) is processed through three convolutional layers to compress spectral bands, thereby obtaining a

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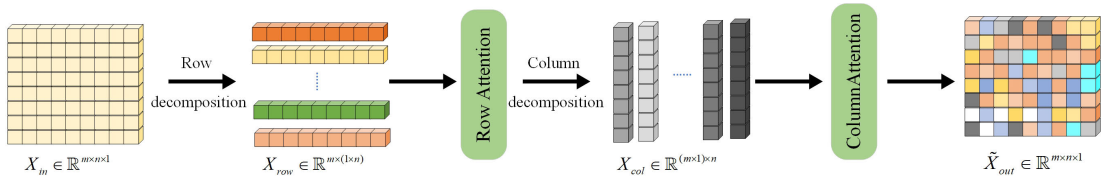


Fig. 2. Overall architecture of AA. In AA, the feature maps sequentially undergo processing by row attention followed by column attention, thereby extracting global feature information.

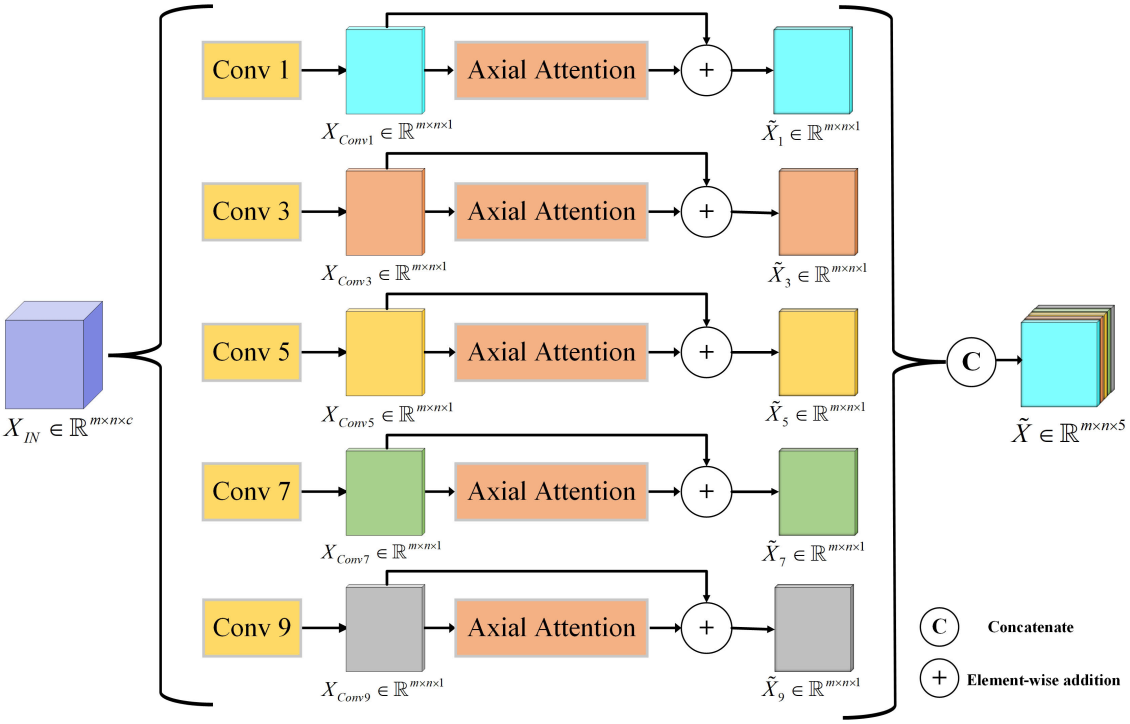


Fig. 3. Overall architecture of MAA. In MAA, the input feature maps undergo integration via residual connections incorporating multiscale convolutions and AA mechanisms, ultimately yielding multiscale local–global feature information through concatenation.

more refined spatial representation. The specific formula is represented as follows:

$$\mathrm{lr}_{\mathrm{down}} = \mathrm{Relu}(\mathrm{Con}v_3(\mathrm{Relu}(\mathrm{Con}v_3(\mathrm{Relu}(\mathrm{Con}v_3(\mathrm{lr})))))) \quad (9)$$

where $\mathrm{Con}v_3(\cdot)$, denotes the operation of a CNN with a 3×3 convolution kernel, and $\mathrm{Relu}(\cdot)$, indicates the ReLU activation function

$$\mathrm{lr}_{\mathrm{spatial}} = \mathrm{Relu}(\mathrm{Con}v_3(F_{\mathrm{MAA}}(\mathrm{lr}_{\mathrm{down}}))) \quad (10)$$

where $\mathrm{lr}_{\mathrm{down}} \in \mathbb{R}^{h\times w\times L//8}$ undergoes the MAA Module $F_{\mathrm{MAA}}(\cdot)$, resulting in the output $\mathrm{lr}_{\mathrm{spatial}} \in \mathbb{R}^{h\times w\times 5}$, effectively extracting spatial information from LR-HSI.

B. Spectral-FE-HSI

Extracting spectral information from HSIs poses challenges due to high-dimensional data and various interferences. To address these issues, we have designed a spectral-FE module, whose specific process is illustrated in Fig. 1(b).

1) Depthwise Self-Attention: Fig. 4 illustrates the overall architecture of the DW-Atten module. The DW-Atten module is designed to efficiently and intricately capture interactions between individual bands in hyperspectral and multispectral images. This can be mathematically represented as

$$X_{\mathrm{OUT}} = F_{\mathrm{Selfatten}}(F_{\mathrm{GAP}}(\mathrm{Con}v_{\mathrm{DW}}(X_{\mathrm{IN}}))) \otimes X_{\mathrm{IN}} \quad (11)$$

where the input feature map $X_{\mathrm{IN}} \in \mathbb{R}^{m\times n\times C}$ undergoes a Depthwise convolution $\mathrm{Con}v_{\mathrm{DW}}(\cdot)$, to produce the output $X_{\mathrm{DW}} \in \mathbb{R}^{m\times n\times C}$. X_{DW} undergoes Global Average Pooling $F_{\mathrm{GAP}}(\cdot)$, and is then multiplied element-wise $\otimes$ with the input X_{IN} after passing through the self-attention mechanism $F_{\mathrm{Selfatten}}(\cdot)$, to obtain the output result $X_{\mathrm{OUT}} \in \mathbb{R}^{m\times n\times C}$.

2) Spectral Feature Extraction: The primary goal of the Spectral-FE module's design is to extract spectral information from LR-HSI, shown in Fig. 1(b). The specific formula is represented as follows:

$$\mathrm{lr}_{\mathrm{spectral}} = F_{\mathrm{DW}_{\mathrm{atten}}}(F_{\mathrm{re}}(\mathrm{FC}_{\mathrm{re}}(\mathrm{AA}(\mathrm{FC}(F_{\mathrm{Flatten}}(\mathrm{lr})))))) \quad (12)$$

where the flattening operation $F_{\mathrm{Flatten}}(\cdot)$, is applied to the spatial dimensions of $\mathrm{lr} \in \mathbb{R}^{h\times w\times L}$, followed by dimensionality reduction in the spectral dimension through the fully connected layer $\mathrm{FC}(\cdot)$. Subsequently, axial attention $\mathrm{AA}(\cdot)$ is utilized to emphasize key spectral features. Then, the previously compressed spectral dimension is reconstructed using a reverse fully connected layer $\mathrm{FC}_{\mathrm{re}}(\cdot)$ to restore and

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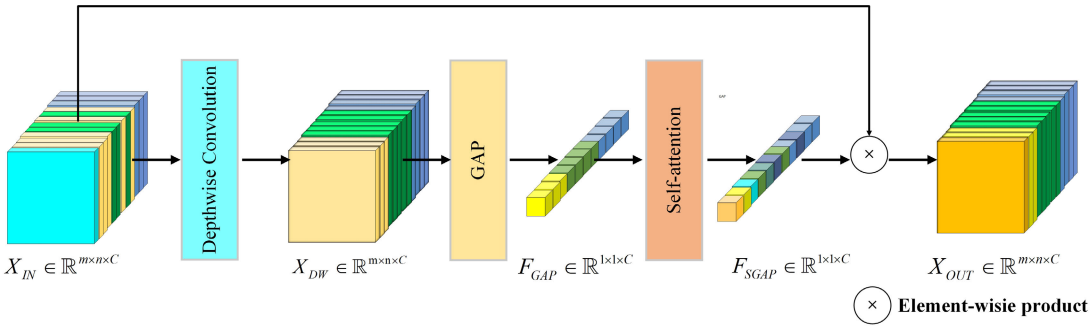


Fig. 4. Overall architecture of DW-Atten. Feature maps, subjected to depthwise convolution, global average pooling, and self-attention mechanisms, yield channel weights, which are then used to weight and derive spectral features.

maintain feature richness. After reshaping via $F_{\text{re}}(\cdot)$, the spatial–spectral structure of the features is restored, followed by spectral self-attention using depthwise self-attention $F_{\text{DW}_{\text{atten}}}(\cdot)$, to obtain $\text{lr}_{\text{spectral}} \in \mathbb{R}^{h\times w\times L}$.

C. Feature Injection Module

As shown in Fig. 1(c), the design of the FIM aims to fully leverage the higher spatial resolution and finer spatial information of MSI. Through this module, our focus is on extracting multiscale local–global spatial features of MSI to ensure the integration of rich and comprehensive information in the reconstruction process of HR-HSI. FIM leverages the MAA module's capabilities to extract multiscale local–global feature information from HR-MSI ($\text{hr} \in \mathbb{R}^{H\times W\times 5}$), enhancing spatial resolution and detail. FIM also includes various levels of hierarchical downsampling, strategically injected into the HR-HSI reconstruction process to enhance the representation of objects in high-resolution contexts. The process can be elucidated as follows:

$$\tilde{X}_{\text{hr}} = F_{\text{MAA}}(\text{hr}) \tag{13}$$

where hr is first passed through $F_{\text{MAA}}(\cdot)$, to obtain multiscale local–global features $\tilde{X}_{\text{hr}} \in \mathbb{R}^{H\times W\times 5}$. Subsequently, $\tilde{X}_{\text{lr}}$ is downsampled to varying degrees to match each fusion stage. The specific formulas are as follows:

$$\tilde{X}^{\text{hr}}_{\text{low}} = F_{\text{Avg}}\left(F_{\text{Avg}}\left(\text{Relu}\left(\text{Con}v_3\left(\tilde{X}_{\text{hr}}\right)\right)\right)\right) \tag{14}$$

where $\tilde{X}_{\text{hr}}$ undergoes two layers of average pooling $F_{\text{Avg}}(\cdot)$, after passing through the convolutional layer $\text{Con}v_3(\cdot)$, and the $\text{Relu}(\cdot)$, activation function, resulting in feature information with low spatial resolution $\tilde{X}^{\text{hr}}_{\text{low}} \in \mathbb{R}^{H/4\times W/4\times 5}$. This allows $\tilde{X}^{\text{hr}}_{\text{low}}$ to better adapt to the initial interaction stage of features

$$\tilde{X}^{\text{hr}}_{\text{med}} = F_{\text{Avg}}\left(\text{Relu}\left(\left(\text{Con}v_3\left(\tilde{X}_{\text{hr}}\right)\right)\right)\right) \tag{15}$$

similar to the process of obtaining $\tilde{X}^{\text{hr}}_{\text{low}}$, the acquisition of medium-resolution feature information $\tilde{X}^{\text{hr}}_{\text{med}} \in \mathbb{R}^{H/2\times W/2\times 5}$ only involves a single application of average pooling. $\tilde{X}^{\text{hr}}_{\text{med}}$ is better suited for the intermediate feature interaction stage

$$\tilde{X}^{\text{hr}}_{\text{hig}} = \text{Relu}\left(\text{Con}v_3\left(\tilde{X}_{\text{hr}}\right)\right) \tag{16}$$

where $\tilde{X}_{\text{hr}}$ undergoes only convolutional operations without reducing the spatial resolution to obtain $\tilde{X}^{\text{hr}}_{\text{hig}} \in \mathbb{R}^{H\times W\times 5}$. This allows $\tilde{X}^{\text{hr}}_{\text{hig}}$ to retain more spatial feature information, enabling it to better suit the final feature interaction stage.

D. Progressive Fusion Module

As shown in Fig. 1(d), the PFM is designed to reconstruct HR-HSI using features extracted from previous steps progressively. It works in a hierarchical and progressive manner, where features are first extracted through convolutional fusion and then upsampled layer by layer using transposed convolution to enhance spatial resolution. During reconstruction, the PFM incorporates multiscale features extracted by the FIM, capturing spatial structures and details at various scales and providing rich spatial context. The PFM employs a progressive strategy, gradually enhancing feature resolution through transposed convolution layers, thereby continuously optimizing the reconstructed output. This approach not only recovers the high-frequency details of the HSI but also integrates multiscale features, significantly enhancing the spatial accuracy and structural integrity of the reconstructed image. Its formula can be represented as follows:

$$\text{lr}_{\text{ss}} = \text{Relu}\left(\text{Con}v_3\left(\text{Concat}\left\{\text{lr}_{\text{spatial}}, \text{lr}_{\text{spectral}}\right\}\right)\right) \tag{17}$$

where the $\text{lr}_{\text{spatial}} \in \mathbb{R}^{h\times w\times 5}$ and $\text{lr}_{\text{spectral}} \in \mathbb{R}^{h\times w\times L}$ extracted from the Spatial-FE and Spectral-FE modules are first concatenated along the channel dimension. Subsequently, the concatenated features undergo fusion through $\text{Con}v_3(\cdot)$, followed by $\text{Relu}(\cdot)$, activation function, resulting in the formation of $\text{lr}_{\text{ss}} \in \mathbb{R}^{h\times w\times L}$

$$\text{fu}_{\text{low}} = \text{Relu}\left(\text{Conv}_3\left(\text{Concat}\left\{\tilde{X}^{\text{hr}}_{\text{low}}, \text{lr}_{\text{ss}}\right\}\right)\right) \tag{18}$$

where after concatenating lr_{ss} and $\tilde{X}^{\text{hr}}_{\text{low}}$, the fused low-resolution feature fu_{low} is obtained through $\text{Con}v_3(\cdot)$, and $\text{Relu}(\cdot)$, activation function. Subsequently, the $\text{fu}_{\text{low}} \in \mathbb{R}^{h\times w\times L}$ is mapped to the med-resolution fusion result $\hat{H}_{\text{med}} \in \mathbb{R}^{H/2\times W/2\times L}$ through transposed convolution, for which the formula is as follows:

$$\hat{H}_{\text{med}} = \text{Transpose}(\text{fu}_{\text{low}}) \tag{19}$$

where $\text{Transpose}(\cdot)$, denotes the transpose convolution operation. The low-resolution features are mapped to the medium-resolution fusion result in $\hat{H}_{\text{med}}$ and then fused with $\tilde{X}^{\text{hr}}_{\text{med}}$ using a convolutional layer, reducing channels to L

$$\text{fu}_{\text{med}} = \text{Relu}\left(\text{Con}v_3\left(\text{Concat}\left\{\tilde{X}^{\text{hr}}_{\text{med}}, \hat{H}_{\text{med}}\right\}\right)\right) \tag{20}$$

where after fusion, $\tilde{X}^{\text{hr}}_{\text{med}}$ and $\hat{H}_{\text{med}}$ produce a medium-resolution feature $\text{fu}_{\text{med}} \in \mathbb{R}^{W/2\times H/2\times L}$ that

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incorporates medium-resolution spatial feature information, and this is then mapped to a high-resolution fusion result

$$\hat{H}_{\text{hig}} = \text{Transpose}(\text{fu}_{\text{med}}) \tag{21}$$

where the medium-resolution feature fu_{med} is mapped to the high-resolution fusion result $\hat{H}_{\text{hig}} \in \mathbb{R}^{H\times W\times L}$ through transpose convolution

$$\hat{H} = \text{Relu}\left(\text{Con}\nu_3\left(\text{Concat}\left\{\tilde{X}^{\text{hr}}_{\text{hig}}, \hat{H}_{\text{hig}}\right\}\right)\right) \tag{22}$$

where $\hat{H} \in \mathbb{R}^{H\times W\times L}$ indicates the fused HR-HSI; where the high-resolution fusion result $\hat{H}_{\text{hig}}$ is further fused with the high-resolution spatial feature $\tilde{X}^{\text{hr}}_{\text{hig}}$ to correct the final fusion result, and the spectral dimension is mapped to a specified size. The steps of the proposed PMA^2-Net algorithm are outlined in Algorithm 1.

Algorithm 1 PMA^2-Net

Input: High-resolution multispectral images (hr), Low-resolution hyperspectral images (lr)

1: Extract multiscale local–global information $\tilde{X}_{hr}$ from hr via Eq. 13.
2: Obtain downsampling features of different sizes $\tilde{X}^{hr}_{low}$, $\tilde{X}^{hr}_{med}$, $\tilde{X}^{hr}_{hig}$ via Eq. 14, 15, and 16.
3: Extraction of lr spatial information: Obtain lr_{down} via Eq. 9, followed by deriving lr spatial feature information $lr_{spatial}$ via Eq. 10.
4: Extraction of lr spectral information: Obtain $lr_{spectral}$ via Eq. 12.
5: Concatenate and fuse $lr_{spatial}$ and $lr_{spectral}$ via Eq. 17 to obtain lr_{ss}.
6: Concatenate $\tilde{X}^{hr}_{low}$ and lr_{ss}, perform fusion to obtain fu_{low} via Eq. 18, and upsampling to obtain $\hat{H}_{med}$ via Eq. 19.
7: Concatenate $\tilde{X}^{hr}_{med}$ and $\hat{H}_{med}$, perform fusion to obtain fu_{med} via Eq. 20, and upsampling to obtain $\hat{H}_{hig}$ via Eq. 21.
8: Finally, perform feature fusion of $\tilde{X}^{hr}_{hig}$ and $\hat{H}_{hig}$ via Eq. 22.

Output: High-Resolution Hyperspectral Images ($\hat{H}$)

E. Loss Function

In this study, we trained hyperspectral–multispectral fusion networks using mean squared error (mse), a straightforward but commonly used loss function. The definition of the loss function L_{mse} is as follows:

$$L_{\text{mse}} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{L}\left(\left(\hat{H}(i,j,k) - H(i,j,k)\right)^2\right)}{2HWL} \tag{23}$$

where $\hat{H} \in \mathbb{R}^{H\times W\times L}$ denotes the reconstructed HR-HSI, and $H \in \mathbb{R}^{H\times W\times L}$ represents the reference HR-HSI.

IV. Experiment Results and Analysis

A. Datasets Description

1) Botswana: The Botswana dataset was acquired over the Okavango Delta in Botswana by the Hyperion sensor on NASA's EO-1 satellite from 2001 to 2004. There are 242 bands in all in Botswana, spanning a spectral range of 0.4 to 2.5 μm at intervals of 10 nm.

2) Pavia Center: The ROSIS sensor was used to collect the Pavia Center dataset, which included a 115-band original dataset. 102 bands were chosen for this investigation after variables like noise and water vapor were eliminated.

3) Pavia University: The University of Pavia dataset was similarly collected with the ROSIS sensor, same like the Pavia Center dataset. We used 103 of the 115 bands that this sensor could record after processing.

4) Houston: The Houston dataset was obtained from the 2013 IEEE GRSS Data Fusion Contest and was obtained with the ITRES CASI-1500 sensor. With a spatial resolution of 349 $\times$ 1905, this dataset contains 144 bands and a spectral range of 364 to 1046 nm.

5) Cuprite: The Cuprite dataset, collected in 1997 using the AVIRIS sensor, covers the Cuprite area in Nevada, USA. It has a spatial resolution of 20 m and 512 $\times$ 614 pixels with 224 bands ranging in wavelength from 370 to 2480 nm.

B. Evaluation Metrics

To quantitatively evaluate the performance of different models, this study introduces four evaluation metrics: peak signal-to-noise ratio (PSNR), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), SAM, and structural similarity index measure (SSIM).

1) Peak Signal-to-Noise Ratio: PSNR is a metric that compares the quality of fused or reconstructed images to the original images. The ratio of the signal's peak power to the noise power added during the reconstruction or fusion process quantifies the quality. A higher PSNR value indicates better image quality with less distortion. The formula for computing PSNR in the cth band is defined as

$$\text{PSNR}\left(x^c, y^c\right) = 10\log_{10}\left(\frac{\max(y^c)^2}{\frac{1}{H*W}\|x^c - y^c\|_2^2}\right) \tag{24}$$

where $\max(\cdot)$ is a function that returns the maximum value. $x^c \in \mathbb{R}^{H\times W\times 1}$ and $y^c \in \mathbb{R}^{H\times W\times 1}$ represent the cth band images of the fused image and the reference image, respectively. $||\cdot||_2$ refers to the two-norm. H and W denote the number of rows and columns of the image. For a HSI with C bands, the final PSNR value is the average of the PSNR values of each band.

2) Erreur Relative Globale Adimensionelle de Synthèse: ERGAS is a global metric that comprehensively considers the overall performance of images, making it particularly useful for evaluating the effectiveness of image fusion algorithms. ERGAS is commonly used to assess the performance of image fusion algorithms, particularly in hyperspectral imaging. A smaller ERGAS value indicates better fusion quality. The calculation formula is as follows:

$$\text{ERGAS}(x, y) = \frac{100}{r}\sqrt{\frac{1}{C}\sum_{c=1}^{C}\frac{\|y^c - x^c\|_2^2}{\mu^2(y^c)}} \tag{25}$$

where r is the downsampling ratio, and $\mu(\cdot)$ denotes the mean value.

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3) Spectral Angle Mapper: SAM is a method used to assess the spectral accuracy of image fusion results. Its main goal is to measure the spectral angle between the fused image and the reference image to evaluate the spectral precision of the fusion results. The following formula can be used to calculate SAM:

$$\mathrm{SAM}(x, y) = \frac{1}{H * W} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \arccos\left(\frac{\left\langle x_{(i,j)} \bullet y_{(i,j)} \right\rangle}{\left\| x_{(i,j)} \right\|_2 \left\| y_{(i,j)} \right\|_2} \right) \tag{26}$$

where $x_{(i,j)} \in \mathbb{R}^{1\times1\times\mathbb{C}}$ and $y_{(i,j)} \in \mathbb{R}^{1\times1\times\mathbb{C}}$ represent the spectral vectors at position (i, j) in the two images, and $\langle\bullet\rangle$ denotes the inner product operation.

4) Structure Similarity Index Measure: SSIM is a metric used to evaluate image quality and is widely applied in the assessment of image fusion. SSIM considers not only the similarity between pixels but also the global structural information of the image. This enables SSIM to capture the overall quality of images when evaluating fusion results

$$\mathrm{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{27}$$

where μ_x and μ_y denote the mean luminance values of the two images, with C_1 as a stability constant. The standard deviations of the two images are represented by σ_x and σ_y, respectively, and C_2 is another stability constant. The covariance between the two images is denoted by σ_{xy}, with C_3 as a third stability constant.

C. Implementation Details

For all five datasets, two input images are generated according to the protocol outlined by Wald [24]. The original HSI acts as the reference image for HR-HSI (denoted as H). LR-HSI (denoted as lr) is defined as the input, obtained by downsampling the blurred HR-HSI, which is preprocessed using a 5×5 Gaussian filter with a standard deviation of 3 and a downsampling ratio (r) of 4. The input HR-MSI (denoted as hr) consists of five images evenly spaced across the HR-HSI

$$\begin{aligned} \mathrm{lr} &= \mathrm{Bilinear}(\mathrm{Gaussian}(H), 1/r) \\ \mathrm{hr}_k &= H\left(\left\lfloor \frac{(k-1)\cdot L}{l-1} \right\rfloor\right), \quad k \in \{1, 2, \ldots, l\} \end{aligned} \tag{28}$$

where the lr undergoes Gaussian filtering by H for blur processing and subsequently undergoes a bilinear operation to achieve downsampling by a factor of r in the spatial dimension. The hr bands are selected from H at uniform intervals, and subregions are extracted as test images while the remaining areas are used for training, where hr_k represents the kth band and $\lfloor\cdot\rfloor$ denotes the floor operation. Random cropping is performed on training images with the same spatial resolution (128×128) taken from the training region. To avoid overlap between training and testing samples, zero-padding is applied to test regions during model training [18]. For traditional methods, parameters are set to be identical to the original literature, except for data processing. Deep learning experiments are implemented using TensorFlow 2.0.0 in Python 3.7. The computing setup includes a 32-GB CPU and $1 \times$ GPU GeForce RTX 3070. The Adam optimizer is used, with a batch size of 3, 1000 iterations, and a learning rate of $1e^{-4}$.

TABLE I
EXPERIMENTAL RESULTS ON THE BOTSWANA DATASET. RED: THE BEST, BLUE: THE SECOND-BEST

Method	Botswana			
	PSNR	ERGAS	SAM	SSIM
CNMF	23.4455	9.9951	15.0530	0.7176
LTTR	29.0964	5.4094	7.3120	0.4295
MSDCNN	31.7594	9.5550	1.8131	0.9919
ResTFNet	33.2591	1.7715	1.6488	0.9985
TFNet	33.8567	2.2208	1.4438	0.9988
SSFCNN	27.7329	10.2666	2.9977	0.9852
ConSSFCNN	24.4449	10.7680	3.8268	0.9809
SSR-NET	33.4131	1.6260	1.6495	0.9986
MCT-Net	35.6206	1.4135	1.2782	0.9991
MSST-Net	32.6598	1.8579	2.8095	0.9746
MIMO-SST	33.9440	3.7140	1.5379	0.9977
PMA^2-Net	36.5042	1.2588	1.1816	0.9993

D. Comparison With SOTA Methods

We used five classical hyperspectral datasets to compare the PMA^2-Net approach with 11 SOTA methods. These include two traditional fusion methods (CNMF [33], LTTR [45]), six deep learning-based methods using CNNs (MSDCNN [38], TFNet [39], ResTFNet [39], SSFCNN [37], ConSSFCNN [37], SSR-NET [18]), and three transformer-based deep learning methods (MCT-NET [19], MSST-NET [22], MIMO-SST [44]). The visual fusion results are displayed in Figs. 5–9, and the performance metrics of these fusion algorithms for different datasets are displayed in Tables I–V. For comprehensive comparisons, we provide RGB images of the fusion results and offer enlarged detailed images, as well as SAM images depicting the fusion results against reference images. Due to space constraints, only the outcomes of deep learning-based fusion techniques are displayed.

1) Results on Botswana: Table I and Fig. 5 present the objective evaluation metrics and visual comparisons on the Botswana dataset between PMA^2-Net and other fusion methods. PMA^2-Net achieves excellent visual fusion results with minimal overall spectral variation in the global SAM plot. It attains the best scores across all metrics, indicating satisfactory outcomes. Despite the low spatial resolution of the Botswana dataset, PMA^2-Net delivers satisfactory results, demonstrating its proficiency in spatial information reconstruction in such scenarios. PMA^2-Net surpasses MCT-NET in PSNR by 0.88 dB, highlighting its ability to retain information. Furthermore, the visual results reveal sharper image edges following PMA^2-Net fusion.

2) Results on Pavia Center: Table II presents the objective evaluation metrics for all compared methods on the Pavia Center dataset, while Fig. 6 shows the visual results. According to the experimental findings, PMA^2-Net performs well in the PSNR and ERGAS metrics. Compared to MSST-Net, PMA^2-Net shows a 3.27 dB increase in PSNR. Additionally, it shows a 0.93 reduction in SAM compared

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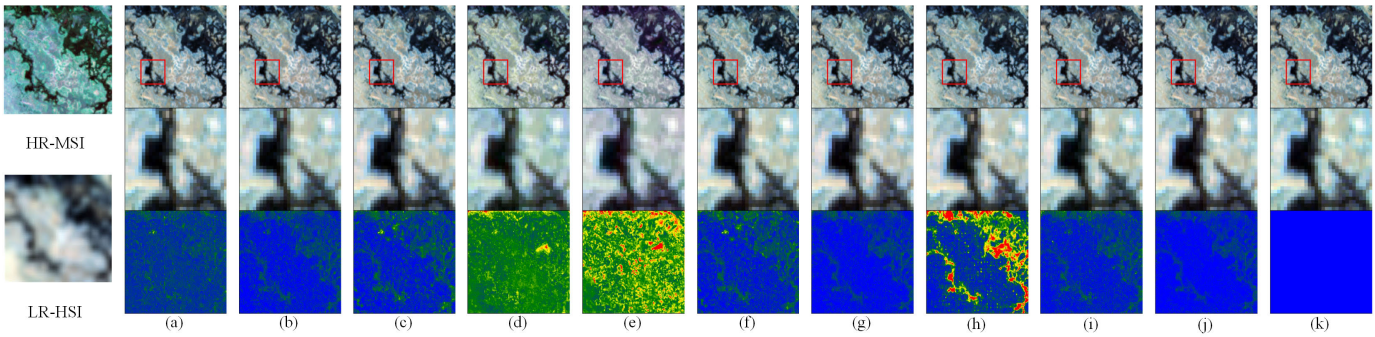


Fig. 5. Fusion results of different methods on Botswana dataset. The corresponding input images (HR-MSI and LR-HSI) are on the far left. The first row shows the R-G-B images (100-70-40 bands) of the estimated HR-HSIs, the second row shows the locally enlarged image of the R-G-B image, and the third row shows the SAM between the estimated and the reference HR-HSIs. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) MCT-Net. (h) MSST-Net. (i) MIMO-SST. (j) PMA^2-Net. (k) Ground truth.

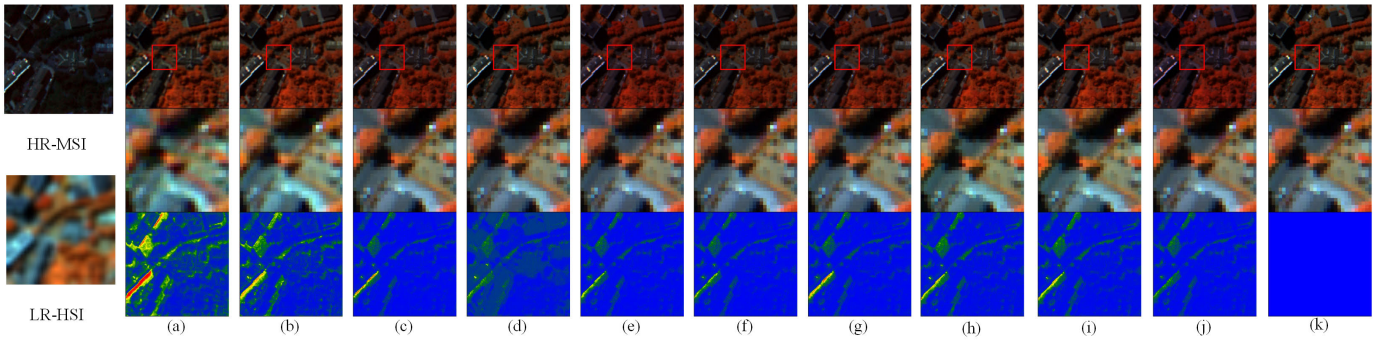


Fig. 6. Fusion results of different methods on Pavia Centre dataset. The corresponding input images (HR-MSI, LR-HSI) are on the far left. The first row shows the R-G-B images (100-70-40 bands) of the estimated HR-HSIs, the second row shows the locally enlarged image of the R-G-B image, and the third row shows the SAM between the estimated and the reference HR-HSIs. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) MCT-Net. (h) MSST-Net. (i) MIMO-SST. (j) PMA^2-Net. (k) Ground truth.

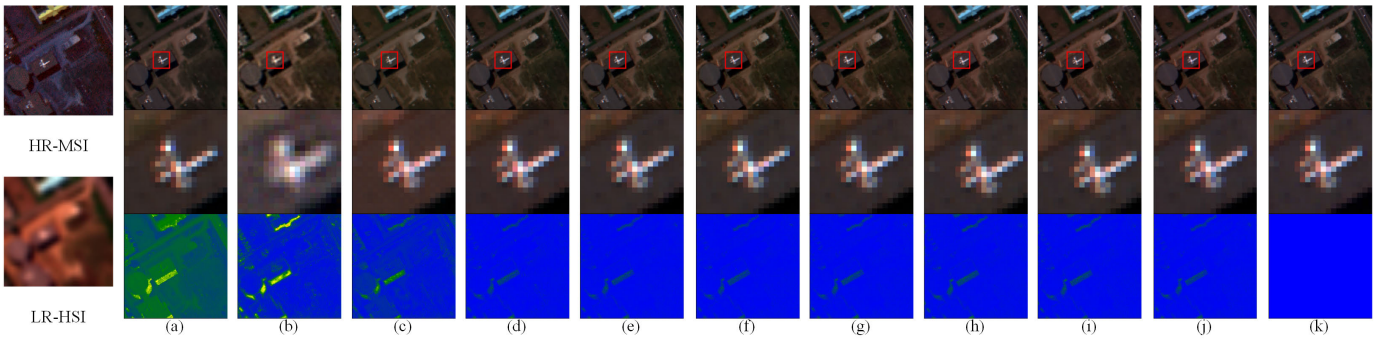


Fig. 7. Fusion results of different methods on Pavia University dataset. The corresponding input images (HR-MSI, LR-HSI) are on the far left. The first row shows the R-G-B images (60-30-10 bands) of the estimated HR-HSIs, the second row shows the locally enlarged image of the R-G-B image, and the third row shows the SAM between the estimated and the reference HR-HSIs. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) MCT-Net. (h) MSST-Net. (i) MIMO-SST. (j) PMA^2-Net. (k) Ground truth.

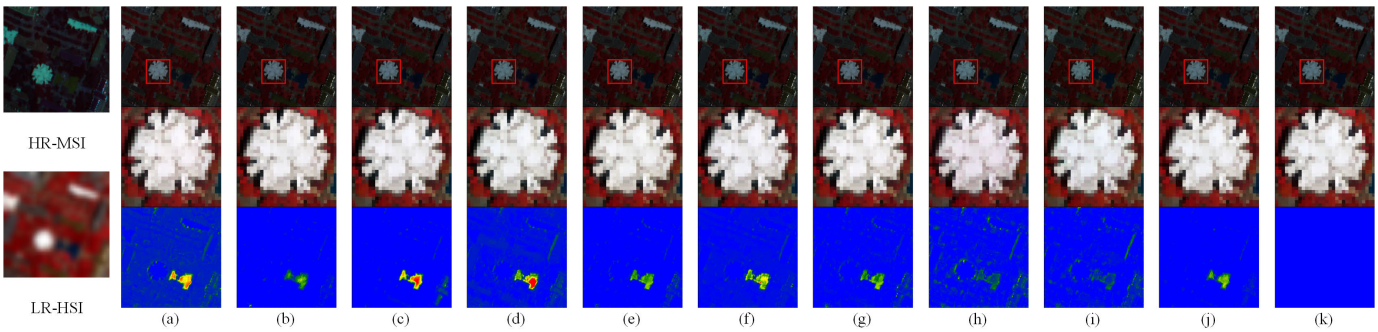


Fig. 8. Fusion results of different methods on Houston dataset. The corresponding input images (HR-MSI, LR-HSI) are on the far left. The first row shows the R-G-B images (100-70-40 bands) of the estimated HR-HSIs, the second row shows the locally enlarged image of the R-G-B image, and the third row shows the SAM between the estimated and the reference HR-HSIs. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) MCT-Net. (h) MSST-Net. (i) MIMO-SST. (j) PMA^2-Net. (k) Ground truth.

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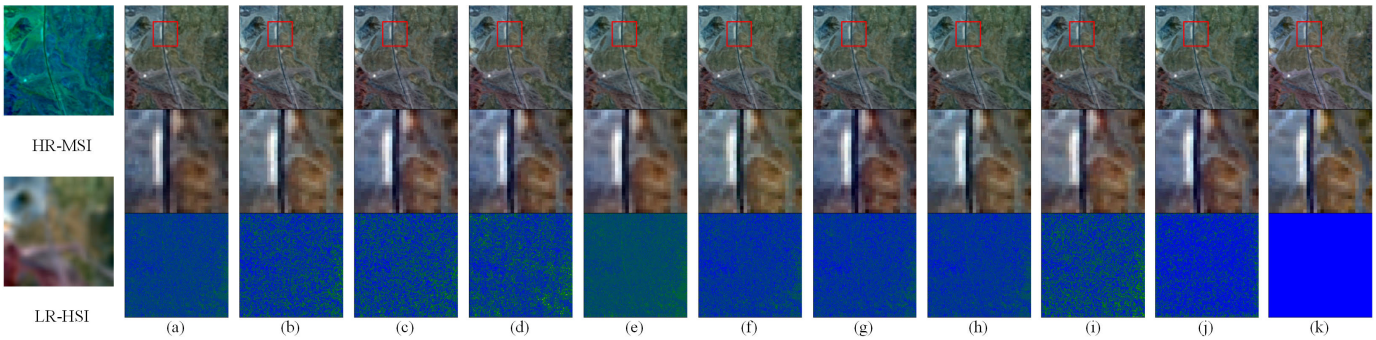


Fig. 9. Fusion results of different methods on Cuprite dataset. The corresponding input images (HR-MSI, LR-HSI) are on the far left. The first row shows the R-G-B images (100-70-40 bands) of the estimated HR-HSIs, the second row shows the locally enlarged image of the R-G-B image, and the third row shows the SAM between the estimated and the reference HR-HSIs. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) MCT-Net. (h) MSST-Net. (i) MIMO-SST. (j) PMA^2-Net. (k) Ground truth.

TABLE II

EXPERIMENTAL RESULTS ON THE PAVIA CENTER. RED: THE BEST, BLUE: THE SECOND-BEST

Method	Pavia Center			
	PSNR	ERGAS	SAM	SSIM
CNMF	25.2260	8.1491	11.2620	0.8052
LTTR	19.7582	14.9517	29.5117	0.4295
MSDCNN	30.5127	8.0017	10.3996	0.8797
ResTFNet	39.8946	3.1072	3.4385	0.9871
TFNet	33.2549	5.8987	7.0502	0.9434
SSFCNN	40.7002	3.0076	3.3352	0.9871
ConSSFCNN	39.2873	6.4293	6.2407	0.9663
SSR-NET	40.2167	3.1057	3.2872	0.9875
MCT-Net	39.9607	3.2575	3.6066	0.9862
MSST-Net	37.7429	4.0328	4.4056	0.9787
MIMO-SST	38.4360	3.5705	4.2494	0.9816
PMA^2-Net	41.0090	2.8944	3.3204	0.9868

TABLE III

EXPERIMENTAL RESULTS ON THE PAVIA UNIVERSITY DATASET. RED: THE BEST, BLUE: THE SECOND-BEST

Method	Pavia University			
	PSNR	ERGAS	SAM	SSIM
CNMF	31.9792	3.5486	5.2879	0.9626
LTTR	27.2151	6.4211	9.6441	0.8914
MSDCNN	31.3791	8.5889	13.6213	0.8776
ResTFNet	32.6211	3.1156	4.7170	0.9405
TFNet	30.2927	3.9971	5.5748	0.8664
SSFCNN	43.4782	1.1969	1.7831	0.9855
ConSSFCNN	42.5840	1.3011	2.0718	0.9820
SSR-NET	44.2647	1.1534	1.7933	0.9858
MCT-Net	43.9914	1.2036	1.9237	0.9853
MSST-Net	41.6709	1.3806	2.2027	0.9831
MIMO-SST	43.0246	1.2568	1.9850	0.9847
PMA^2-Net	45.1965	1.1300	1.7744	0.9862

to MIMO-SST, demonstrating its spectral solid information reconstruction capability. Our PMA^2-Net also achieves satisfactory visual reconstruction results with minimal spectral variation.

3) Results on Pavia University: While Fig. 7 displays the visualization outcomes of all fusion approaches, Table III presents the assessment metrics for each model on the Pavia University dataset. The experimental results show that PMA^2-Net outperforms other comparative models across multiple evaluation metrics, delivering excellent results in both spatial and spectral reconstructions. Despite the high spatial resolution of 1.3 m in the Pavia University dataset, PMA^2-Net achieves outstanding visual results. Compared to direct one-time upsampling, progressive upsampling exploits the relationship between LR and HR images, allowing for more effective spectral information injection.

4) Results on Houston: Table IV presents the evaluation metrics for fusion results on the Houston dataset, while Fig. 8 shows the visual comparisons. Experimental results demonstrate that PMA^2-Net outperforms MCT-Net in all four metrics, achieving a notable PSNR improvement of 1.53 dB and an ERGAS below 1, highlighting its superior spectral reconstruction capability. Visually, PMA^2-Net produces sharper HSI images with minimal spectral distortion and enhanced color fidelity, underscoring its effectiveness in spatial structure preservation and high-quality image fusion.

TABLE IV

EXPERIMENTAL RESULTS ON THE HOUSTON DATASET. RED: THE BEST, BLUE: THE SECOND-BEST

Method	Houston			
	PSNR	ERGAS	SAM	SSIM
CNMF	36.5051	2.2277	3.5821	0.9863
LTTR	31.8523	3.6763	5.2879	0.9607
MSDCNN	41.1217	5.1257	3.2892	0.9718
ResTFNet	46.1509	1.2474	1.3787	0.9959
TFNet	44.3112	1.3359	1.2120	0.9970
SSFCNN	44.1944	5.0676	2.6091	0.9729
ConSSFCNN	46.7882	1.2395	1.4123	0.9970
SSR-NET	46.1614	4.5397	2.2541	0.9781
MCT-Net	44.9986	1.4750	1.8616	0.9961
MSST-Net	42.1580	1.8896	2.2974	0.9937
MIMO-SST	44.0246	1.9249	1.9958	0.9950
PMA^2-Net	48.3217	0.9718	1.1733	0.9973

5) Results on Cuprite: Fig. 9 shows the fusion results of each approach, and Table V lists the metrics used to evaluate each fusion method on the Cuprite dataset. The experiments utilized the Cuprite dataset, known for its diverse terrain, and the findings show that PMA^2-Net outperformed all four criteria. Despite the challenging terrain conditions and the dataset's 224 spectral bands, PMA^2-Net achieved the best SAM indicator, demonstrating its exceptional ability to reconstruct spectral information and its robustness in handling tasks that require high spectral resolution.

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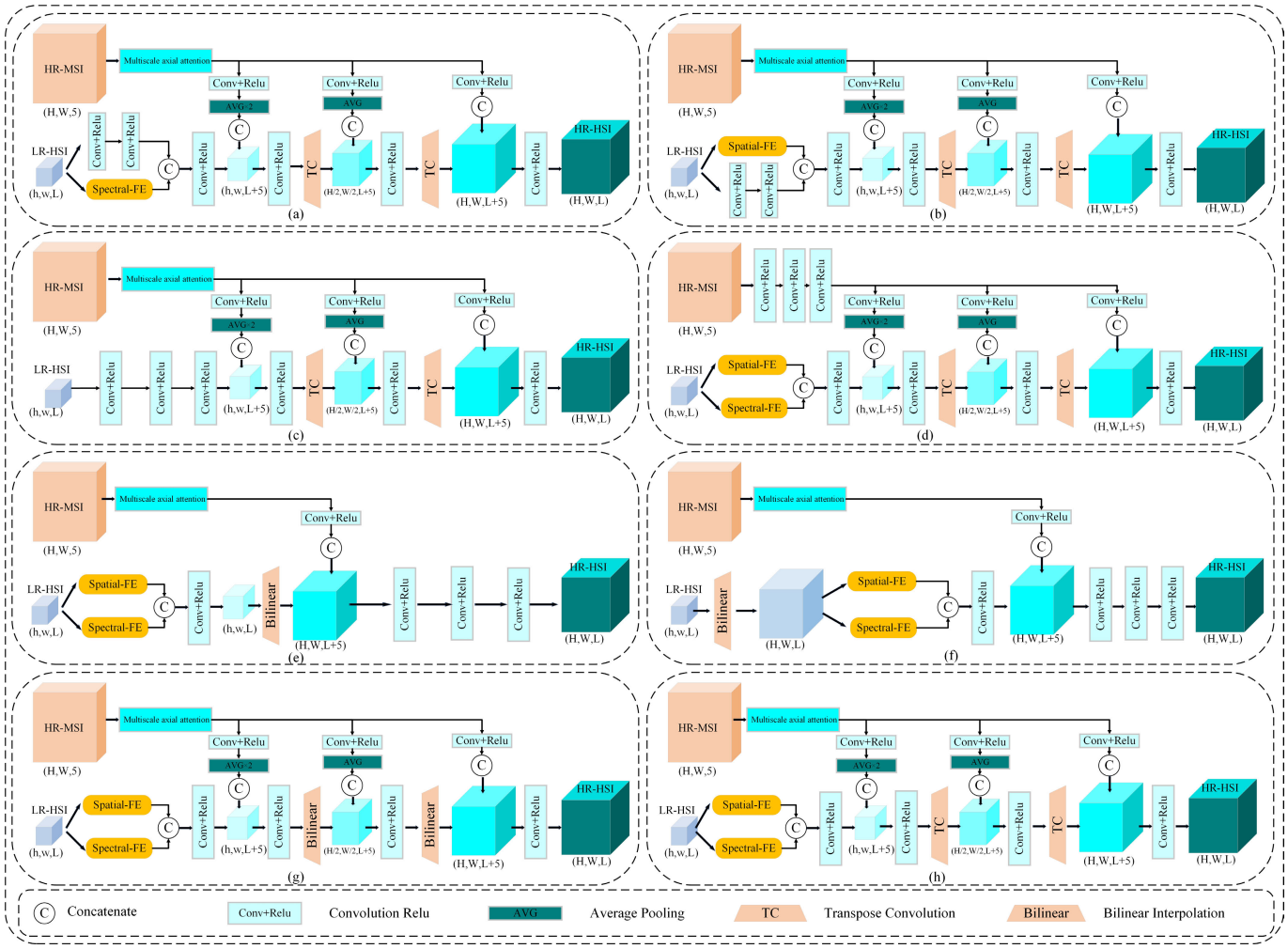


Fig. 10. Network architecture diagrams for each ablation experiment. (a) Replacing Spatial-FE with convolution. (b) Replacing Spectral-FE with convolution. (c) Replacing Spatial-FE and Spectral-FE with convolution. (d) Replacing FIM with convolution. (e) Replacing PFM with convolution. (f) Upsampling. (g) Upsampling uses Bilinear. (h) PMA^2-Net.

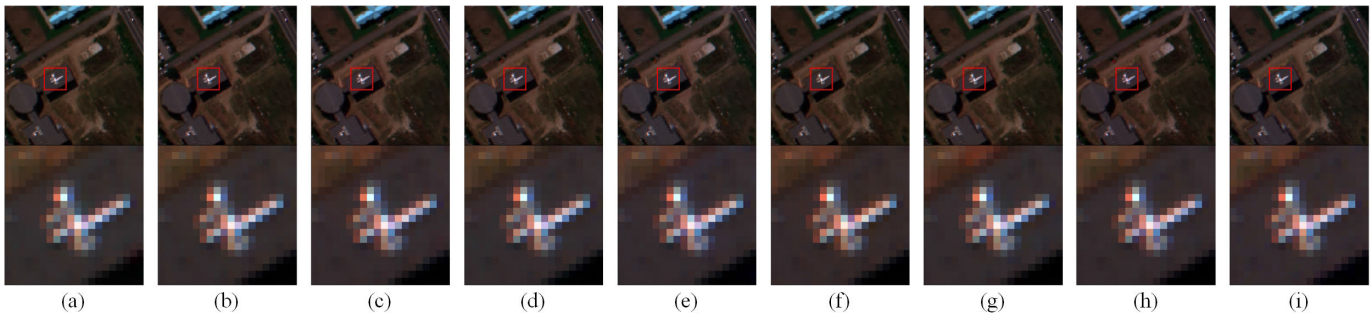


Fig. 11. Fusion results of ablation study. The first row shows the R-G-B images (60-30-10 bands) of the estimated HR-HSIs, and the second row shows the locally enlarged R-G-B image. (a) Replacing Spatial-FE with convolution. (b) Replacing Spectral-FE with convolution. (c) Replacing Spatial-FE and Spectral-FE with convolution. (d) Replacing FIM with convolution. (e) Replacing PFM with convolution. (f) Upsampling. (g) Upsampling uses Bilinear. (h) PMA^2-Net. (i) Ground truth.

E. Ablation Analysis

To evaluate the effectiveness of each module in PMA^2-Net, we conducted a systematic series of ablation experiments using the Pavia University dataset. These experiments are divided into three sections: the first focuses on analyzing the Spatial-FE and Spectral-FE modules to assess their effectiveness in extracting features from LR-HSI; the second investigates the role of the FIM in extracting multiscale local–global spatial information; and the third examines the contribution of the PFM to detail feature recovery via ablation, as well as the impact of hierarchical reconstruction on fusion results. Fig. 10 shows the network architecture used in the experiments, Fig. 11 provides specific visualization results, and Fig. 12 depicts the SAM visualizations. Additionally,

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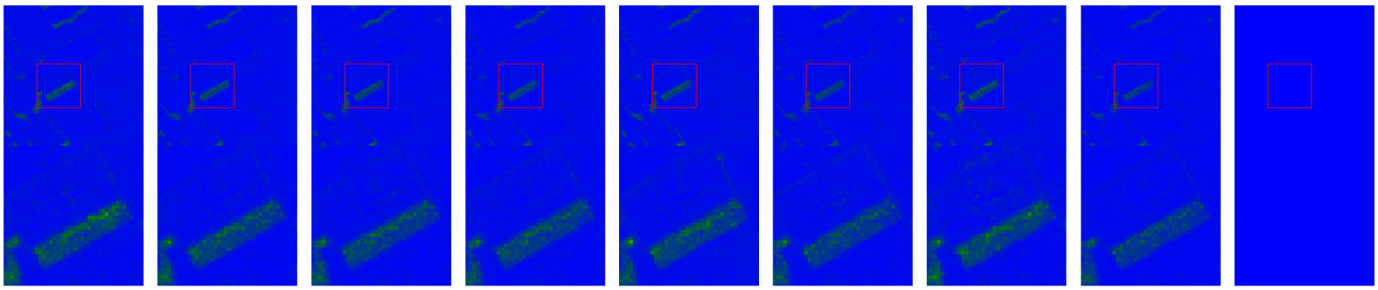


Fig. 12. SAM results of ablation study. The first row displays the SAM calculated between the fusion results and the reference HR-HSIs, while the second row shows the corresponding locally enlarged SAM. (a) Replacing Spatial-FE with convolution. (b) Replacing Spectral-FE with convolution. (c) Replacing Spatial-FE and Spectral-FE with convolution. (d) Replacing FIM with convolution. (e) Replacing PFM with convolution. (f) Upsampling. (g) Upsampling uses Bilinear. (h) PMA^2-Net. (i) Ground truth.

TABLE V
EXPERIMENTAL RESULTS ON THE CUPRITE DATASET. RED: THE BEST, BLUE: THE SECOND-BEST

Method	Cuprite			
	PSNR	ERGAS	SAM	SSIM
CNMF	27.6016	7.4247	13.2365	0.8838
LTTR	22.9457	9.5585	11.9346	0.6297
MSDCNN	32.6966	8.2999	9.1456	0.9480
ResTFNet	35.6684	3.6770	9.1113	0.9717
TFNet	36.5297	3.6189	8.7596	0.9724
SSFCNN	33.3630	8.2886	9.3791	0.9582
ConSSFCNN	31.0499	9.5796	11.4185	0.9201
SSR-NET	33.3065	8.9617	7.7848	0.9607
MCT-Net	34.0005	7.5047	7.7403	0.9600
MSST-Net	32.3487	8.6108	7.7563	0.9611
MIMO-SST	34.2374	3.7236	9.3310	0.9682
PMA^2-Net	37.3802	3.0854	5.9366	0.9788

Table VI lists several key evaluation metrics, including PSNR, SSIM, ERGAS, and SAM, alongside model parameters and testing times, to comprehensively assess the network's performance under different experimental conditions.

1) Ablation Experiments on Spatial-FE and Spectral-FE: To validate the effectiveness of these modules, three ablation experiments were conducted: substituting Spatial-FE and Spectral-FE individually and replacing both with conventional layers concurrently. Fig. 10(a)–(c) illustrates the architectural adjustments, and Table VI summarizes the outcomes. The unmodified network design surpassed the modified ones in all four evaluation metrics. The results of the ablation experiments show that replacing the Spatial-FE alone, while reducing the testing time, significantly degraded performance, specifically resulting in a PSNR decrease of 1.18 dB, an ERGAS increase of 0.22, a SAM rise of 0.31, and an SSIM decrease of 0.01. When replacing Spectral-FE, the PSNR reduction increases to 16.58 dB, accompanied by concurrent increments in SAM and ERGAS, with no change in testing time. Conversely, the joint replacement of both modules with traditional convolutional layers leads to a severe performance decline, with a PSNR drop of 1.44 dB and a SAM increase of 0.25. Notably, testing time triples, solidifying the pivotal role of Spatial-FE and Spectral-FE modules in augmenting model performance and operational efficiency.

2) Ablation Experiments on the FIM: We substituted the FIM with conventional layers [see Fig. 10(d)], leading to a reduction in model parameters and testing time. However, this significantly degraded performance: PSNR decreased by 0.28 dB, ERGAS increased by 0.34, SSIM decreased by 0.01, and SAM increased by 0.56. Additionally, more severe local spectral distortions are evident in Figs. 11(d) and 12(d). This ablation study powerfully illustrates that the MAA employed by the FIM offers substantial benefits. It effectively retrieves multiscale local–global information from MSI, significantly boosting the signal-to-noise ratio and SSIM of the fused images. Additionally, adopting a stepwise injection approach regulates spectral distortion and detail precision, significantly improving the quality of the resultant fused images.

3) Ablation Experiments on the PFM: To validate the effectiveness of the PFM, we conducted two ablation experiments. First, the PFM was removed, as shown in Fig. 10(e) and (f). Second, we compared the upsampling performance of bilinear interpolation with that of transpose convolution, as illustrated by the structural diagrams in Fig. 10(g) and (h). The experiments demonstrated that removing the PFM and using bilinear interpolation for upsampling resulted in a significant decrease in PSNR. As shown in Fig. 10(f), direct upsampling followed by feature fusion caused a 2.33 dB drop in PSNR, highlighting substantial spatial information loss. From the comparisons in Fig. 12(e)–(h), we observe that the PFM effectively reduces both local and global spectral distortions, resulting in the reconstruction of higher quality images. These ablation experiments underscore the effectiveness of the PFM in progressively enhancing spatial resolution through multiscale feature fusion and upsampling. Further analysis of upsampling methods revealed that although bilinear interpolation reduces the parameter count by 0.1912 M and accelerates inference speed, it also causes a 3.30 dB decline in PSNR and worsens other evaluation metrics. In contrast, transpose convolution exhibits superior learning capacity during upsampling by leveraging task-specific attributes and achieving end-to-end optimization through close integration with convolutional layers. While bilinear interpolation is simple and fast, it fails to generate new features and relies solely on pixel interpolation. Therefore, transpose convolution holds a clear advantage in scenarios requiring the model to adjust dynamically based on context due to its superior learning capability and enhanced upsampling results.

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TABLE VI

RESULTS OF ABLATION EXPERIMENTS ON THE PAVIA UNIVERSITY DATASET. RED: THE BEST, BLUE: THE SECOND-BEST

FIM	Spatial-FE	Spectral-FE	PFM	TransposeConv	BILINEAR	Params (M)	Test time (Ms)	PSNR	ERGAS	SAM	SSIM
√		√	√	√		1.2936	77.9996	44.0194	1.3486	2.0805	0.9809
√	√		√	√		1.2265	101.0008	43.5607	1.2897	2.0426	0.9820
√			√	√		1.1795	398.0014	43.7569	1.2816	2.0194	0.9821
	√	√	√	√		0.6543	84.0010	42.3802	1.4746	2.3310	0.9783
√	√	√		√		1.1400	98.0005	42.7816	1.4074	2.2511	0.9808
	√	√		√		0.9170	75.0012	42.8710	1.2982	2.0511	0.9833
√	√	√	√		√	0.9584	97.9996	41.8985	1.5221	2.3642	0.9766
√	√	√	√	√		1.1496	101.001	45.1965	1.1300	1.7744	0.9862

TABLE VII

PARAMETERS AND TEST TIME OF THE PROPOSED PMA^2-NET AND THE OTHER DEEP LEARNING MODELS. THE UNITS FOR PARAMETERS ARE IN MILLIONS, AND THE UNITS FOR TEST TIME ARE IN MILLISECONDS. RED: THE BEST, BLUE: THE SECOND-BEST

Method	Pavia University			
	Params	Test time	PSNR	SSIM
MSDCNN	1.2522	12.9995	31.3791	0.8776
ResTFNet	2.3249	34.0006	32.6211	0.9405
TFNet	2.4496	33.0002	30.2927	0.8664
SSFCNN	0.4611	4.0004	43.4782	0.9855
ConSSFCNN	0.4786	10.0007	42.5840	0.9820
SSR-NET	0.2867	21.0032	44.2647	0.9858
MCT-Net	4.5468	77.0495	43.9914	0.9853
MSST-Net	42.9915	1188.7210	41.6709	0.9831
MIMO-SST	5.1210	478.7880	43.0246	0.9847
PMA^2-Net	1.1496	101.0010	45.1965	0.9864

F. Analysis of Time Efficiency

Table VII presents each method's parameters and processing time applied to the Pavia University dataset. Moreover, we include the PSNR and SSIM metrics for each approach to enable rigorous comparisons. CNN-based models (MSDCNN [38], ResTFNet [39], TFNet [39], SSFCNN [37], ConSSFCNN [37], and SSR-NET [18]) have the advantage of faster inference speeds. However, their image reconstruction performance declines due to CNNs' limitations in global feature extraction. In contrast, PMA^2-Net excels at extracting both local and global features and improves image reconstruction through a progressive reconstruction strategy. Compared to Transformer-based deep learning models (MCT-Net [19], MSST-Net [22], and MIMO-SST [44]), PMA^2-Net achieves better fusion results with fewer parameters. This is largely attributed to the AA mechanism, which offers lower computational complexity compared to traditional self-attention mechanisms while maintaining high performance. The MAA module integrates multiscale local and global feature information with minimal parameter overhead and network depth. Through progressive upsampling, extracted features integrate seamlessly into the HR-HSI reconstruction process. However, existing models are constrained by fluctuations in hyperspectral band quantities, as they depend on a fixed number of bands. Mismatches necessitate model retraining, severely impeding model scalability and applicability. Future endeavors will address this limitation, exploring expedited fusion architectures to achieve a judicious balance between performance and efficiency.

V. CONCLUSION

In this article, we propose a novel PMA^2-Net for HSI and MSI fusion. The PMA^2-Net consists of four modules: Spatial-FE, Spectral-FE, FIM, and PFM. Extensive experiments on five hyperspectral datasets demonstrate the superiority and effectiveness of PMA^2-Net. Although CNN-based models (such as MSDCNN and SSR-NET) offer faster inference speeds, their limited ability to extract global features results in less-than-ideal image reconstruction performance. In contrast, PMA^2-Net utilizes its FIM and PFM to capture both local and global features efficiently, significantly enhancing image reconstruction quality. Additionally, compared to Transformer-based models like MSST-Net and MIMO-SST, PMA^2-Net achieves better fusion results with fewer parameters due to its AA mechanism, which reduces computational complexity while maintaining high performance.

Comprehensive ablation studies further validate the critical role of each module. Spatial-FE and Spectral-FE modules are essential for boosting model performance. FIM effectively enhances image quality through multiscale feature extraction, and its removal causes noticeable spectral distortion. PFM excels in progressively enhancing spatial resolution, with transposed convolutions outperforming bilinear interpolation during upsampling, enabling higher quality feature generation and end-to-end optimization. Furthermore, we anticipate potential applications of this approach in various image fusion tasks, particularly in remote sensing image processing.

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